

DEEPAKRAJ B

BACKEND SOFTWARE ENGINEER | JAVA, KOTLIN, SPRING BOOT & AWS

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PROFILE

Backend Software Engineer with hands-on experience in a product-based company environment, building scalable event-driven microservices, real-time telemetry streaming pipelines, and enterprise platform integrations. Proven track record taking complex operational requirements - from on-site field constraints at an Army camp to large-scale commercial fleet telemetry for IOCL - and translating them into reliable, high-throughput production systems. Focused on robust data consistency, clean architectural boundaries, and pragmatic engineering that keeps mission-critical platforms dependable under real-world loads.

PROFESSIONAL EXPERIENCE

Software Engineer | Kritilabs Technologies Limited, Chennai, India July 2024 - Present

IOCL Journey Risk Management (JRM) Platform **Python, Java 21, Spring Boot, Kafka, PostgreSQL, Uber H3, Valhalla**

- Architecting the spatial map-matching and telemetry risk-scoring pipeline for IOCL's Journey Risk Management platform, targeting **150,000+ commercial transit routes**.
- Built a modular Python processing pipeline (**GPS fix-quality filtering, H3 hexagonal indexing, Valhalla map-matching**) to classify routes into Red/Yellow/Green exposure zones, recorded as **ADRs**.
- Reconciled live telemetry event schemas to surface and resolve a **1000x data discrepancy**, designing cross-service event contracts with **Java 21 / Spring Boot & PostgreSQL** for automated operational alerts.

Event-Driven Microservices & Cloud Platform **Java, Spring Boot 3.x, Spring Cloud, Kafka, Spring Data, PostgreSQL, AWS**

- Architected event-driven microservices decomposing monoliths using **Java, Spring Boot 3.x, and Spring Cloud** (Config Server, Eureka, Gateway) with **Apache Kafka** for async inter-service streaming.
- Enforced transactional **data consistency and exactly-once processing (EOS)** across consumer groups, boosting query throughput by **35%** with composite indexing and query tuning on **PostgreSQL & MySQL**.
- Hardened API gateways with dynamic **RBAC via Spring Security & JWT**, integrated **Resilience4j** circuit breakers, and automated database migrations with **Flyway** and **Terraform** on **AWS EKS**.

SentinelKote - Field Asset Tracking & Offline Sync **Java, Spring Boot, REST APIs, Android, Offline Sync, RFID Hardware**

- Deployed on-site at an **Indian Army camp** to architect and deliver an offline-first inventory tracking system, replacing manual paper ledger audits with hardware RFID verification within **2 weeks**.
- Engineered offline synchronization protocols and deterministic conflict-resolution logic, guaranteeing transactional data consistency and zero ledger discrepancies upon network reconnection.
- Built modular Java backend REST services with strict validation boundaries, audit trail logging, and low-latency local storage, ensuring sub-second response times under constrained field conditions.

Applied AI & SDLC Automation Systems **FastAPI, LangGraph, SQLGlott, LangChain, Pydantic, Bitbucket, Redis**

- NL2SQL Reasoning Engine:** Built an **NL2SQL engine** (LangGraph, FastAPI, SQLGlott guardrails) reducing query latency by **35%** with Redis caching, and automated PRD parsing to Redmine backlogs via **LangChain & Pydantic**.
- Automated PR Review Pipeline:** Engineered an automated code review engine triggered in **Bitbucket Pipelines**, parsing git diffs and AST changes via **Python & LLM APIs** with Pydantic validation to enforce architectural rules before merge.

TECHNICAL SKILLS

JVM & Backend: Java (11/17/21), Kotlin, Spring Boot 3.x, Spring Cloud, Spring Data JPA/JDBC

Databases & Spatial: PostgreSQL, MySQL, Redis Caching, Flyway Migrations

Cloud Infrastructure: AWS (EKS, RDS, VPC, S3), Serverless, Terraform, Kubernetes, Docker

APIs & Security: RESTful APIs, Spring Security, JWT, Dynamic RBAC, Resilience4j (Circuit Breaker)

Streaming & Messaging: Apache Kafka, Kafka Streams, Avro Schemas, Exactly-Once Processing (EOS)

Spatial & Telemetry: Uber H3 Grid Indexing, Valhalla Map-Matching, Telemetry Pipelines, ADRs

CI/CD & DevOps: Bitbucket Pipelines, Docker, Docker Compose, Linux, Shell / Bash

Testing & AI Tooling: JUnit 5, Mockito, Testcontainers, Swagger / OpenAPI, LangGraph, SQLGlott, Pytest

EDUCATION

B.Tech in Computer Science & Engineering | SASTRA Deemed University, Thanjavur, India CGPA: 7.30 | 2020 - 2024

CERTIFICATIONS & TRAINING

- Backend Development Specialization - Meta
- Cloud Computing & System Architecture - Coursera
- Software Engineering Specialization - IBM
- Python Crash Course - Google